

Unai Gurbindo Gonzalez

Trustworthy AI | Uncertainty Quantification | Robustness under Distribution Shift

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Research Profile

PhD researcher working on uncertainty quantification and trustworthy AI under distribution shift. My research focuses on *domain uncertainty*, out-of-distribution detection and the limits of predictive uncertainty as tools for robust model evaluation. I also study how uncertainty-aware evaluation connects to responsible AI documentation, accountability and ethical deployment.

Research Interests

Uncertainty quantification and calibration; epistemic, aleatoric and domain uncertainty; out-of-distribution detection; robustness under distribution shift; trustworthy and responsible AI; model documentation, impact assessment and accountable deployment.

Education

PhD in Artificial Intelligence

Universitat Politècnica de Catalunya & Barcelona Supercomputing Center

Nov 2025 – May 2029 expected

Barcelona, Spain

Thesis: *Trust under Shift: Domain Uncertainty Modeling for Trustworthy and Ethical AI*. Supervisors: Axel Brando and Caroline König. FI-STEP Fellowship, awarded Sep 2025.

MSc in Artificial Intelligence

Universitat Politècnica de Catalunya / Universitat de Barcelona / Universitat Rovira i Virgili

Sep 2023 – Jan 2025

Barcelona, Spain

GPA: 9.290/10. Thesis: *Image Generation for Improving the Robustness of Computer Vision Models in Autonomous Driving*. Supervisors: Axel Brando, Caroline König and Jaume Abella.

BSc in Data Science

Public University of Navarre

Sep 2019 – Jun 2023

Navarre, Spain

GPA: 8.949/10. Thesis: *Characterisation of the Immune System in the Progression to ACLF*.

Publications

Accepted / Published

- **Gurbindo Gonzalez, U.***, Aldoma, A.*, & Brando, A. *Position: Predictive Uncertainty Is Not Enough – Joint Distribution for Full Uncertainty Representation*. International Conference on Machine Learning, Position Paper Track, 2026. Accepted. *Equal contribution.
- **Gurbindo, U.**, Brando, A., Abella, J., & König, C. *Object Detection in Adverse Weather Conditions for Autonomous Vehicles Using Instruct Pix2Pix*. International Joint Conference on Neural Networks, 2025, pp. 1–8. [doi:10.1109/IJCNN64981.2025.11228623](https://doi.org/10.1109/IJCNN64981.2025.11228623).
- Palomino, S., Huergo, E., Uson, E., Sanchez, C., Lagani, V., Kiani, N., Planell, N., **Gurbindo, U.**, Trebicka, J., Farias, A., Caraceni, P., Rautou, P.-E., & Gomez-Cabrero, D. *Robust Identification of Patient Subgroups in Acute Decompensated Cirrhosis*. *Journal of Hepatology*, 78, S290, 2023. [doi:10.1016/S0168-8278\(23\)00877-2](https://doi.org/10.1016/S0168-8278(23)00877-2).

Submitted / Under Review

- **Gurbindo Gonzalez, U.**, Brando, A., & Valls, P. *Stop Footnoting Ethics in AI Documentation: Use Care Cards*. Submitted manuscript.

Research Experience

PhD Researcher in Trustworthy AI

Barcelona Supercomputing Center & Universitat Politècnica de Catalunya

Nov 2025 – Present

Barcelona, Spain

AI Institute – TRUE. Supervisor: Axel Brando.

- Investigate uncertainty quantification methods for model evaluation under distribution shift, with emphasis on predictive, epistemic, aleatoric and domain-level uncertainty.
- Develop the notion of *domain uncertainty* to analyse when inputs fall outside the support or assumptions of the deployment domain.
- Study how uncertainty-aware evaluation can inform AI documentation, accountability mechanisms and responsible deployment decisions.

Junior Research Engineer in Robust AI

Barcelona Supercomputing Center

AI Institute – TRUE.

Jan 2025 – Nov 2025

Barcelona, Spain

- Built reproducible pipelines for evaluating uncertainty estimates in deep learning models across ID, OOD and corrupted-data settings.
- Analysed failure modes of predictive uncertainty under distribution shift, including overlap between high label uncertainty and OOD inputs.
- Evaluated deep ensembles on robustness, calibration and uncertainty behaviour using corruption benchmarks and mixed ID/OOD test sets.
- Contributed empirical analysis to work on joint uncertainty representation for trustworthy AI.

Master Thesis Researcher in Autonomous Driving

Barcelona Supercomputing Center

Supervisors: Axel Brando, Caroline König and Jaume Abella.

Jul 2024 – Jan 2025

Barcelona, Spain

- Designed a generative data-augmentation pipeline using InstructPix2Pix to improve object-detection robustness under adverse weather shifts.
- Evaluated YOLOv10 and Faster R-CNN on CARLA, BDD100K and ACDC, analysing robustness gaps between clean and weather-shifted visual domains.
- Contributed to an IJCNN 2025 publication on robust perception for autonomous driving.

Bioinformatics Research Intern and Bachelor Thesis Researcher

Navarrabiomed, Translational Bioinformatics Unit

Biomedical research centre.

Sep 2022 – Jun 2023

Navarre, Spain

- Analysed clinical and single-cell biomedical data to characterise immune-system patterns in acute-on-chronic liver failure.
- Applied statistical analysis and data visualisation methods to patient stratification in a translational bioinformatics setting.

Technical Skills

Programming: Python, R, Bash, SQL, MATLAB

ML Frameworks: PyTorch, scikit-learn, Jupyter, Weights & Biases

Methods: uncertainty quantification, calibration, OOD detection, robustness evaluation, deep ensembles, computer vision

Research Engineering: Git, Linux, Docker, reproducible experimentation, LaTeX

Responsible AI: AI documentation, impact assessment, accountability frameworks, care ethics

Teaching, Talks and Outreach

Instructor, Trustworthy AI

Chamber of Commerce of Terrassa

Delivered applied sessions for local SMEs on trustworthy AI, covering uncertainty, risk assessment, documentation and responsible adoption.

Jan 2026 – Feb 2026

Terrassa, Spain

Invited Speaker, European Researchers' Night

Public University of Navarre

Presented scientific outreach content as part of a public research communication event.

Sep 2022

Navarre, Spain

Participant, iGEM Competition

UPNAvarra_Spain, Public University of Navarre

Participated in an interdisciplinary synthetic biology project involving scientific communication and data-oriented research.

Jun 2021 – Oct 2022

Navarre, Spain

Awards, Funding and Languages

Awards & Funding

- FI-STEP Fellowship, awarded Sep 2025.
- Ranked 1st in class, BSc in Data Science, Public University of Navarre.
- Course-level honours/distinctions: 10 during BSc studies; 2 during MSc studies.

Selected Training: AI for Medicine Specialization, deeplearning.ai / Coursera, 2023.

Languages: Basque – Native; Spanish – Native; English – Professional working proficiency (B2).